

Facial Expression Recognition using Deep Learning (CNN)

1. Introduction

This project focuses on Facial Expression Recognition using Deep Learning and Convolutional Neural Networks (CNN). The system detects human emotions from facial images such as happy, sad, angry, surprise, fear, disgust, and neutral.

2. Project Objectives

The main objective is to build a complete deep learning pipeline including EDA, preprocessing, CNN modeling, training, evaluation, and deployment preparation.

3. Technologies Used

Python, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Seaborn, OpenCV, and Scikit-learn were used in this project.

4. Dataset Description

The dataset contains facial images categorized into multiple emotion classes. The dataset was divided into training, validation, and testing sets.

5. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset structure, image distribution, class balance, and image quality. Visualization techniques and count analysis were used.

6. Data Preprocessing

Images were resized, normalized, and augmented using ImageDataGenerator. Data augmentation included rotation, zooming, shifting, and horizontal flipping.

7. CNN Architecture

The CNN model included Conv2D, MaxPooling2D, Dropout, Flatten, and Dense layers. The architecture was designed to extract facial features and classify emotions accurately.

8. Model Compilation

The model was compiled using Adam optimizer and categorical crossentropy loss function with accuracy as the evaluation metric.

9. Model Training

The model was trained for 20 epochs using training and validation datasets. Training accuracy increased while loss decreased during the learning process.

10. Accuracy and Loss Visualization

Training and validation graphs were generated using Matplotlib to visualize accuracy and loss trends across epochs.

11. Model Evaluation

The trained model was evaluated using validation and test data. Metrics such as accuracy, precision, recall, F1-score, and confusion matrix were considered.

12. Prediction System

The trained model can predict emotions from unseen images by preprocessing the image and passing it through the CNN model.

13. Model Saving

The trained model was saved in .h5 format for future use and deployment.

14. Future Improvements

Future work includes increasing epochs, using transfer learning, hyperparameter tuning, real-time webcam integration, and Streamlit deployment.

15. Applications

Facial expression recognition can be used in healthcare, surveillance, smart learning systems, driver monitoring, customer analysis, and human-computer interaction.

16. Conclusion

This project successfully demonstrated the complete workflow of a facial expression recognition system using CNN and deep learning techniques.

Project Completion Summary

Component	Status
EDA	Completed
Data Preprocessing	Completed
CNN Model Building	Completed
Training (20 Epochs)	Completed
Evaluation	Completed
Model Saving	Completed